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Winter, 2025

B. Tech 4<sup>th</sup> Semester Examination

Hydraulic Engineering

Course Code: BCE23404T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

### SECTION 1

Answer all the questions

Marks: 10 x 1 = 10

1. (a) Laminar flow typically occurs when Reynolds number is:
  - i. Less than 2000
  - ii. Between 2000 and 4000
  - iii. Greater than 4000
  - iv. Equal to zero
- (b) Turbulent flow is characterized by:
  - i. Smooth, orderly layers of fluid
  - ii. Low energy dissipation
  - iii. Random and chaotic fluctuations
  - iv. Absence of velocity gradients
- (c) In fully developed laminar flow through a circular pipe, the velocity profile is:
  - i. Uniform
  - ii. Parabolic
  - iii. Linear
  - iv. Irregular
- (d) The thickness of the boundary layer increases with:
  - i. Distance from the leading edge
  - ii. Decreasing Reynolds number
  - iii. Both i and ii
  - iv. None of the above

(e) The boundary layer is defined as:

- i. A layer where temperature remains constant
- ii. A thin region where fluid viscosity can be neglected
- iii. A region where the flow velocity changes from zero to free-stream value
- iv. A region of turbulent wake formation

(f) The Darcy-Weisbach equation is used to calculate:

- i. Reynolds number
- ii. Friction factor in turbulent flow
- iii. Head loss due to friction
- iv. Boundary layer thickness

(g) Which of the following represents minor losses in pipe flow?

- i. Loss due to wall shear
- ii. Loss due to entrance and exit effects
- iii. Loss due to pipe length
- iv. Loss due to laminar profile

(h) When a jet strikes a stationary vertical flat plate normally, the force exerted by the jet is proportional to:

- i. Velocity
- ii. Velocity<sup>2</sup>
- iii. Area
- iv. Density only

(i) When a jet strikes a curved vane moving in the same direction as the jet, the relative velocity of jet with respect to the vane is:

- i.  $V + u$
- ii.  $u - V$
- iii.  $V - u$
- iv.  $V$

(j) A pump is used to:

- i. Convert kinetic energy into hydraulic energy
- ii. Lift water from low to high level
- iii. Generate electricity
- iv. Reduce pressure in pipes

$V =$

## SECTION 2

Marks: 10

Answer any one question from this section

- 2(a) Derive an expression for the velocity distribution for viscous flow between parallel plates. Also sketch the velocity distribution across a section of the plates. (10)

OR

- 2(b) Determine the wall shearing stress in a pipe of diameter 10cm which carries water. The velocities at the pipe centre and 30mm from the pipe centre are 3m/s and 2m/s respectively. The flow in pipe is given as turbulent. (10)

## SECTION 3

Marks: 10

Answer any one question from this section

- 3(a) For the velocity profile for laminar boundary layer flows given as:

$$\frac{u}{U} = 2 \left( \frac{y}{\delta} \right) - 2 \left( \frac{y}{\delta} \right)^3 + \left( \frac{y}{\delta} \right)^4$$

Determine the boundary layer thickness, shear stress, drag force and co-efficient of drag in terms of Reynold number. (10)

OR

- 3(b) For the velocity profile for laminar boundary layer flows given as:

$$\frac{u}{U} = \frac{3}{2} \left( \frac{y}{\delta} \right) - \frac{1}{2} \left( \frac{y}{\delta} \right)^3$$

Determine the boundary layer thickness, shear stress and co-efficient of drag in terms of Reynold number. (10)

## SECTION 4

Marks: 10

Answer any one question from this section

- 4 (a) A pipe of diameter 300mm and length 3500m is used for the transmission of power by water. The total head at the inlet of the pipe is 500m. Find the maximum power available at the outlet of the pipe, if the value of  $f=0.006$ . (10)

OR

- 4 (b) Find an expression for the power transmission through pipes. What is the condition for maximum transmission of power and corresponding efficiency of transmission? (10)

## SECTION 5

Marks: 10

Answer any one question from this section

5 (a) A jet of water of diameter 10cm strikes a flat plate normally with a velocity of 15m/s. The plate is moving with a velocity of 6m/s in the direction of the jet and away from the jet. Find: (10)

- i. The force exerted by the jet on the plate
- ii. Work done by the jet on the plate per second.
- iii. Power and efficiency of the jet.

OR

(5x5=10)

8 (b) Write a short notes on:

- i. Francis turbine
- ii. Centrifugal pump